

DANIEL T. LACOSTA

Las Piñas City, NCR | danielacostatenorio@gmail.com | 09054336553

LinkedIn: www.linkedin.com/in/danieltenoriolacosta | Portfolio: <https://mendo03.github.io/>

PROJECT SUMMARY

Computer Engineer graduate with practical experience in embedded systems, hardware-software troubleshooting, and IoT prototyping. Internship done in R & D for the development and debugging of Raspberry Pi-based systems using GSM Modules and serial communication. Comfortable working across hardware, software, and networking layers and ready to get absorbed into a technical role from day one.

WORK EXPERIENCE

Research and Development Intern

IoT Technology, Incorporated | 2026

- Provided troubleshooting and technical support for hardware and software systems, identifying and resolving system errors
- Assisted in configuring and testing Raspberry Pi-based systems and communication modules (GSM module, serial interfaces)
- Performed system diagnostics and validation to ensure stable operation of prototypes
- Documented technical procedures, issues, and resolutions to support development and team collaboration

KEY ACCOMPLISHMENTS

SMART VOLTAGE LEAK DETECTION SYSTEM (January 2024)

- Diagnosed and resolved hardware-software integration issues to improve system stability
- Implemented automated fault detection and response mechanisms
- Performed system testing and validation to ensure consistent operation

WATER OVERFLOWING ALARM (June 2024)

- Built and tested a real-time monitoring system using sensors and Arduino
- Troubleshoot sensor inaccuracies and circuit issues during development

- Assisted in PCB integration and system setup

PHILIPPINE ROBOTICS CUP (October 2025)

- Collaborated with a technical team to support event operations and system setup
- Assisted in troubleshooting technical issues during robotics activities

DEVELOPMENT OF SUBMARINE FOR MONITORING AND DETECTING BLOCKAGE IN THE WATERWAYS OF LAS PIÑAS CITY (November 2025)

- Troubleshoot communication system issues between the Raspberry Pi, sensors, and camera modules
- Improved system reliability through debugging and performance optimization
- Assisted in integrating object detection (YOLOv8) for identifying blockages
- Conducted real-world testing in low-visibility environments

RIVER LEVEL DETECTION PROJECT (November 2025)

- Served as a technical consultant for a thesis-based river level detection system project
- Guided Arduino programming, SIM module integration, serial communication, SMS functionality, and power system selection.
- Assisted in troubleshooting hardware and embedded system issues to improve device performance and reliability

SKILLS

IT Support & Troubleshooting

- Hardware and software troubleshooting (desktops, embedded systems)
- System diagnostics and issue resolution
- Basic user support and technical guidance

Operating Systems & Tools

- Windows OS (basic troubleshooting and configuration)
- Linux (Raspberry Pi terminal-based)
- Command Line (Windows CMD, Linux CLI)

Networking

- Basic networking: IP addressing, device connectivity, troubleshooting
- Ethernet cable termination and setup

Technologies

- Raspberry Pi, Arduino
- GSM modules, Serial Communication
- Basic exposure to IoT systems and device integration

Programming

- Python (basic scripting, OpenCV exposure)
- C++ (Arduino-based systems)

EDUCATION

STI College Las Piñas | Bachelor of Science in Computer Engineering

SEMINARS, TRAININGS:

- IoT Conference Philippines 2025
- Automation and Mechatronics in Power Systems Webinar 2025
- Perspective of an ASEAN Engineer on AI in Engineering Education Webinar 2025
- STI EXPERIENCE Artificial Intelligence (AI) Technology Today 2024

- STI XCITE IPv6 Seminar 2024
- STI XCITE Cyber Security Seminar 2024

CERTIFICATIONS

- ISC2 Candidate Certification (Ongoing)
- AWS Academy Cloud Foundation Training Badge (2025)
- Automation and Mechatronics in Power Systems Webinar Certification (2025)
- Perspective of an ASEAN Engineer on AI in Engineering Education Webinar Certification (2025)
- CISCO Computer Hardware Basics Certification (2026)
- CISCO Ethical Hacker Certification (2026)
- CISCO Introduction to Cybersecurity Certification (2026)